

Quiz 1-B

First and Last Name: _____

You have 10 minutes to complete the quiz. Calculators, notes, and other resources are **not** allowed.

1. (2 points) Fill in the elementary row operations in each box.

$$\begin{bmatrix} 1 & 5 & 0 & 2 \\ 2 & 8 & 2 & 2 \\ 0 & 0 & 4 & 9 \end{bmatrix} \xrightarrow{\boxed{}} \begin{bmatrix} 1 & 5 & 0 & 2 \\ 0 & -2 & 2 & -2 \\ 0 & 0 & 4 & 9 \end{bmatrix} \xrightarrow{\boxed{}} \begin{bmatrix} 1 & 5 & 0 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 4 & 9 \end{bmatrix}$$

Solution:

$$\begin{bmatrix} 1 & 5 & 0 & 2 \\ 2 & 8 & 2 & 2 \\ 0 & 0 & 4 & 9 \end{bmatrix} \xrightarrow{r_2 - 2r_1 \rightarrow r_2} \begin{bmatrix} 1 & 5 & 0 & 2 \\ 0 & -2 & 2 & -2 \\ 0 & 0 & 4 & 9 \end{bmatrix} \xrightarrow{r_2 / (-2) \rightarrow r_2} \begin{bmatrix} 1 & 5 & 0 & 2 \\ 0 & 1 & -1 & 1 \\ 0 & 0 & 4 & 9 \end{bmatrix}.$$

QUESTION 2 IS ON THE BACK!!

2. Consider the linear system

$$\begin{cases} x + 2y - 2z = -15 \\ 4x + 2z = 22 \\ 2x + y - z = -3 \end{cases}$$

- (a) (2 points) Write the linear system as an augmented matrix. (Hint: your augmented matrix should have 3 rows, 4 columns)

Solution: The augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 2 & -2 & -15 \\ 4 & 0 & 2 & 22 \\ 2 & 1 & -1 & -3 \end{array} \right].$$

- (b) (2 points) Find the value of r so that $x = 3, y = -4, z = r$ is a solution to the system. Show all work.

Solution: Plug $x = 3, z = r$ into $4x + 2z = 22$, we have $12 + 2r = 22$, then we solve $r = 5$. You can verify this is the solution to the system.